

Task 1

We want to solve the regularized least-squares problem: $\min_x f(x) := \|Ax - b\|_2^2 + \lambda^2 \|x\|_2^2$.

Expanding and differentiating it gives

$$f(x) = (Ax - b)^T(Ax - b) + \lambda^2 x^T x = x^T A^T A x - 2x^T A^T b + b^T b + \lambda^2 x^T x.$$

f is strictly convex, so x minimizes f if and only if $\nabla f(x) = 2(A^T A + \lambda^2 I)x - 2A^T b = 0$, that is, if and only if

$$(A^T A + \lambda^2 I)x = A^T b,$$

which is the normal equation. Since $A^T A + \lambda^2 I$ is positive definite, it is invertible. The normal equations therefore have exactly one solution $x_{\text{RLS}} = (A^T A + \lambda^2 I)^{-1} A^T b$. Which was to be shown.

Now let $A = U\Sigma V^T$ be the SVD of A . Substituting into the normal equations, using that U and V are orthogonal $\Rightarrow U^{-1} = U^T$ and $V^{-1} = V^T$.

$$\begin{aligned} (A^T A + \lambda^2 I)x_{\text{RLS}} &= A^T b \\ V\Sigma^T U^T U \Sigma V^T + \lambda^2 V V^T x_{\text{RLS}} &= V\Sigma^T U^T b \\ V(\Sigma^T \Sigma + \lambda^2 I)V^T x_{\text{RLS}} &= V\Sigma^T U^T b \\ V^T V(\Sigma^T \Sigma + \lambda^2 I)V^T x_{\text{RLS}} &= V^T V \Sigma^T U^T b \\ \underbrace{(\Sigma^T \Sigma + \lambda^2 I)}_{B} \underbrace{V^T x_{\text{RLS}}}_y &= \Sigma^T U^T b \\ B &= \begin{bmatrix} \sigma_1^2 + \lambda^2 & & & \\ & \sigma_2^2 + \lambda^2 & & \\ & & \ddots & \\ & & & \sigma_n^2 + \lambda^2 \end{bmatrix} \\ \Rightarrow y &= B^{-1} \Sigma^T U^T b \\ &= \begin{bmatrix} \frac{1}{\sigma_1^2 + \lambda^2} & & & \\ & \frac{1}{\sigma_2^2 + \lambda^2} & & \\ & & \ddots & \\ & & & \frac{1}{\sigma_n^2 + \lambda^2} \end{bmatrix} \begin{bmatrix} \sigma_1 & & & \\ & \sigma_2 & & \\ & & \ddots & \\ & & & \sigma_n \end{bmatrix} \begin{bmatrix} u_1^T \\ u_2^T \\ \vdots \\ u_n^T \end{bmatrix} b \end{aligned}$$

Since B^{-1} and Σ are diagonal, the i -th component of y is

$$y_i = \frac{\sigma_i}{\sigma_i^2 + \lambda^2} u_i^T b, \quad i = 1, \dots, n.$$

Finally, since $y = V^T x_{\text{RLS}}$ and V is orthogonal, multiplying by V yields $x_{\text{RLS}} = Vy$, which was to be shown.

Since A has full rank, all $\sigma_i > 0$, so we may write $\frac{\sigma_i}{\sigma_i^2 + \lambda^2} = \frac{1}{\sigma_i} \cdot \frac{\sigma_i^2}{\sigma_i^2 + \lambda^2} = \varphi_i \frac{\lambda}{\sigma_i}$, i.e. $y_i = \varphi_i \frac{\lambda}{\sigma_i} u_i^T b$. Expanding $x_{\text{RLS}} = Vy = \sum_i y_i v_i$ then gives

$$x_{\text{RLS}} = Vy = \sum_{i=1}^n \frac{\sigma_i}{\sigma_i^2 + \lambda^2} (u_i^T b) v_i = \sum_{i=1}^n \underbrace{\varphi_i(\lambda)}_{\text{filter factor}} \underbrace{\frac{u_i^T b}{\sigma_i}}_{\text{coefficient of } x_{\text{LS}}} v_i, \quad \varphi_i(\lambda) := \frac{\sigma_i^2}{\sigma_i^2 + \lambda^2}$$

The filter factor $\varphi_i(\lambda)$ acts as a low-pass filter on the terms of the least-squares solution: for $\lambda \ll \sigma_i$ it approaches 1 (term kept), at $\lambda = \sigma_i$ it equals $\frac{1}{2}$, and for $\lambda \gg \sigma_i$ it approaches 0 (term damped).

Task 2

Starting from the (unregularized) least-squares solution expressed via the SVD,

$$x_{\text{LS}} = \sum_{i=1}^r \frac{u_i^T b}{\sigma_i} v_i,$$

small singular values σ_i can amplify noise in b . As in Task 1, we introduce a filter factor $\phi_i(\lambda)$ that damps the contribution of each term:

$$y_i = \frac{\varphi_i(\lambda)}{\sigma_i} u_i^T b, \quad x = Vy = \sum_{i=1}^r \varphi_i(\lambda) \frac{u_i^T b}{\sigma_i} v_i.$$

For the truncated SVD, the filter factor is chosen as

$$\varphi_i = \begin{cases} 1 & \text{if } i \leq k \\ 0 & \text{if } i > k \end{cases} \quad (\sigma_i > \delta) \quad (\sigma_i \leq \delta).$$

Substituting this in, only the first k terms survive:

$$x = \sum_{i=1}^k \frac{u_i^T b}{\sigma_i} v_i + \sum_{i=k+1}^r 0 \cdot \frac{u_i^T b}{\sigma_i} v_i = x_{\text{TLS}}$$

Task 3

We consider the optimization problem

$$\min_x f(x) := \|Ax - b\|_2^2 + \lambda^2 \|Lx\|_2^2$$

By expanding the terms, we get

$$f(x) = (Ax - b)^T(Ax - b) + \lambda^2(Lx)^T(Lx) = x^T A^T A x - 2x^T A^T b + b^T b + \lambda^2 x^T L^T L x.$$

x is minimized when $\nabla f(x) = 2(A^T A + \lambda^2 L^T L)x - 2A^T b = 0$, that is, when

$$(A^T A + \lambda^2 L^T L)x = A^T b$$

This solution is unique when $A^T A + \lambda^2 L^T L$ is invertible.

Task 4

We consider the optimization problem

$$\min_x f(x) := \|Ax - b\|_2^2 + \lambda^2 \|Lx\|_2^2$$

Using GSVD we devide the matrices A and L into the form

$$A = U\Sigma_A W^{-1}, \quad L = V\Sigma_L W^{-1}$$

where

$$\Sigma_A = \begin{bmatrix} \alpha_1 & & & 0 \\ & \alpha_2 & & \\ & & \dots & \vdots \\ 0 & & & \alpha_r \\ & & & 0 \end{bmatrix}, \quad \Sigma_L = \begin{bmatrix} \beta_1 & & & 0 \\ & \beta_2 & & \\ & & \dots & \vdots \\ 0 & & & \beta_r \\ & & & 0 \end{bmatrix}$$

We then expand $f(x)$ and substitute the GSVD of A and L into the normal equations:

$$\begin{aligned} f(x) &= (Ax - b)^T (Ax - b) + \lambda^2 (Lx)^T (Lx) = x^T A^T A x - 2x^T A^T b + b^T b + \lambda^2 x^T L^T L x \\ &= x^T (U\Sigma_A W^{-1})^T U\Sigma_A W^{-1} x - 2x^T (U\Sigma_A W^{-1})^T b + b^T b + \lambda^2 x^T (V\Sigma_L W^{-1})^T V\Sigma_L W^{-1} x \\ &= x^T W^{-1T} \cancel{\Sigma_A^T U^T U} \Sigma_A W^{-1} x - 2x^T W^{-1T} \Sigma_A^T U^T b + b^T b + \lambda^2 x^T W^{-1T} \cancel{\Sigma_L^T V^T V} \Sigma_L W^{-1} x \end{aligned}$$

$$\{y = W^{-1}x\}$$

$$\begin{aligned} &= y^T \Sigma_A^T \Sigma_A y - 2y^T \Sigma_A^T U^T b + b^T b + \lambda^2 y^T \Sigma_L^T \Sigma_L y \\ &= y^T (\Sigma_A^T \Sigma_A + \lambda^2 \Sigma_L^T \Sigma_L) y - 2y^T \Sigma_A^T U^T b + b^T b \end{aligned}$$

differentiating $f(y)$ gives

$$\nabla f(y) = 2(\Sigma_A^T \Sigma_A + \lambda^2 \Sigma_L^T \Sigma_L) y - 2\Sigma_A^T U^T b = 0$$

$$(\Sigma_A^T \Sigma_A + \lambda^2 \Sigma_L^T \Sigma_L) y = \Sigma_A^T U^T b$$

$$B := \Sigma_A^T \Sigma_A + \lambda^2 \Sigma_L^T \Sigma_L = \begin{bmatrix} \alpha_1^2 & & & 0 \\ & \alpha_2^2 & & \\ & & \dots & \vdots \\ 0 & & & \alpha_r^2 \\ & & & 0 \end{bmatrix} + \lambda^2 \begin{bmatrix} \beta_1^2 & & & 0 \\ & \beta_2^2 & & \\ & & \dots & \vdots \\ 0 & & & \beta_r^2 \\ & & & 0 \end{bmatrix} = \begin{bmatrix} \alpha_1^2 + \lambda^2 \beta_1^2 & & & 0 \\ & \alpha_2^2 + \lambda^2 \beta_2^2 & & \\ & & \dots & \vdots \\ 0 & & & \alpha_r^2 + \lambda^2 \beta_r^2 \\ & & & 0 \end{bmatrix}$$

this gives us the final relation

$$y = B^{-1} \Sigma_A^T U^T b = \begin{bmatrix} \frac{1}{\alpha_1^2 + \lambda^2 \beta_1^2} & & & 0 \\ & \frac{1}{\alpha_2^2 + \lambda^2 \beta_2^2} & & \\ & & \dots & \vdots \\ 0 & & & \frac{1}{\alpha_r^2 + \lambda^2 \beta_r^2} \\ & & & 0 \end{bmatrix} \begin{bmatrix} \alpha_1 & & & 0 \\ & \alpha_2 & & \\ & & \dots & \vdots \\ 0 & & & \alpha_r \\ & & & 0 \end{bmatrix} \begin{pmatrix} u_1 \\ \vdots \\ u_r \end{pmatrix}^T b$$

component wise, this gives

$$y_i = \frac{\alpha_i}{\alpha_i^2 + \lambda^2 \beta_i^2} u_i^T b = \underbrace{\varphi_i(\lambda)}_{\text{filter factor}} \underbrace{\frac{u_i^T b}{\alpha_i}}_{\text{coefficient of } x_{\text{LS}}} v_i \quad \varphi_i(\lambda) := \frac{\alpha_i^2}{\alpha_i^2 + \lambda^2 \beta_i^2}$$

Task 5

A samlar in data L vad vi redan vet eller antar om bilden/signalen.

Debluring = göra en suddig bild skarp igen.

$$\int_{\mathbb{R}^d} K(s, t) x(t) dt + e(s) = b(s)$$

t en punkt i skarpa bilden. s en punkt i suddiga bilden.

$K(s, t)$ det som gör bilden suddig. $b(s)$ den suddiga bilden. $e(s)$ brus $x(t)$ den skarpa bilden.

så formeln beskriver från skarp bild till suddig bild. Vi vill göra tvärtom.